

International AS and A-level

Chemistry (9620)

CH02

Report on the Examination

January 2025

REPORT ON THE EXAMINATION: INTERNATIONAL AS CHEMISTRY (9620) CH02 – JANUARY 2025

The paper differentiated well. Students' knowledge of mechanisms, reagents and calculations appears to be improving. Students must use an arrow ($\rightarrow$) not an equal sign ($=$) in chemical reactions to indicate a change. Students must read the question carefully as this was a cause of lost marks.

Other general areas where improvement can be made are as follows:

- Students must show working in calculations.
- Students must clearly cross out any working in drawing formula or calculations that they do not want marked.
- When writing equations, structural formula (if used) must be correctly represented and show the functional groups involved.

QUESTION 01

1.1 Students should know that a line of best fit can also be a curve or a straight line. They should follow that pattern in the data when doing lines of best fit. Lines should go to C18. Many students looked at C8 instead of C18, and/or misread the scale.

1.2 The vast majority of students scored this mark. The most common error was to just state distillation or cracking.

1.3 About half of all students answered this well; the catalyst was well known. The most common errors included not identifying which nitrogen oxide and incorrect balancing of the equation. Students must make sure that carbon monoxide is written clearly as CO not Co.

1.4 The majority of students were able to balance the equation correctly.

1.5 The majority of students were able to balance the equation correctly. Some students were unable to deduce the third product of cracking.

QUESTION 02

2.1 The majority of students answered this question well. The most common error was drawing displayed instead of skeletal formula.

2.2 Students answered this question well but the C=C must be shown for Compound F.

2.3 About half of all students recognised that the spatial arrangement of atoms is needed in the description. The most common error was stating molecular formula rather than structural formula.

2.5 Two thirds of students knew the Z isomer structure. Care must be taken when drawing structures; the correct number of hydrogens per carbon atom needs to be shown.

QUESTION 03

3.1 This was answered well, but many students could not balance the equation or did not know the correct molecular formula for ethene.

3.2 About half of all students answered this correctly. The most common error was not writing an equation or missing the full displayed formula for epoxyethane.

3.3 About half of all students answered this correctly, the silver catalyst was well known.

3.4 Two thirds of students answered this well. A common error was to state water.

QUESTION 04

4.1 About half of the students gained maximum marks. They found mark 2 more challenging. Trailing bonds must come from carbon not hydrogen in polymer structures.

4.2 Despite this question being asked before, less than half of all students stated that non-polar C—C bonds cannot be broken down. Students just defined biodegradable.

4.3 This was answered well by the majority of students. Students need to know how the empirical formula relates to the molecular formula. Where incorrect, either the meaning of empirical formula was not known (so students gave general formula in form of C_nH_{2n+2}) or students rounded the ratio 1:1.5 to 1:2 rather than 2:3.

QUESTION 05

5.1 Students found this question challenging. Over half of all students incorrectly stated that rate was measuring per unit time rather than state it is change in concentration that is measured. The most common error was talking about the speed of a reaction/frequency of collision instead of talking about change/time.

5.2 About half of all students realised the need to heat indirectly due to the flammable nature of ethanol.

5.3 Over a third of students did not recognise the numerical relationship between rate and time. Many students also attempted to name a haloalkane rather than giving one of the options as instructed in the question.

5.4 Two thirds of students correctly answered this question. Students must state C—I bond in their answer. The most common error was to miss out the comparison of carbon—halogen bond strength and discussion of atomic size instead.

5.5 Students realised that fewer particles have the required activation energy; only a third of students discussed collisions in a set time or frequency of successful collisions. The most common error was discussing movement of particles being slower. Many students incorrectly stated that lowering temperature of the reaction lowers the activation energy.

QUESTION 06

6.1 This was answered well by over 80% of students. The most common error was to calculate based on an incorrect number of fluorine atoms in the molecule.

6.2 A third of students answered this well. Students should give a comparative answer when discussing the C–F bond, not just that it is strong or hard to break. Many students did not state that there was no chlorine present in the molecule and stated that M does not contain chlorine radicals.

6.3 This was answered well by two thirds of students. The most common errors were nucleophilic substitution and failing to realise that propagation is a step in free-radical substitution rather than a mechanism.

6.4 This was answered well with three quarters of students achieving maximum marks. The main errors were using an incorrect Mr or rounding too early in the calculation.

6.5 The majority of students answered this well. Students realised smaller volume increases pressure. The most common error was to simply restate the question.

QUESTION 07

7.1 Students answered this question well; common errors were missing out the arrow in M1 and not having the curly arrow in M3 from the lone pair.

7.2 Half of all students recognised that the functional group is to do with chemical reactivity.

7.3 This was answered well by over 50% of students. Common errors included giving incomplete reagents eg OH⁻ and writing (aq) or adding solution to the condition.

7.4 Naming the nitrile was answered well. The most common error was numbering from the wrong end of chain. Students knew KCN but few correctly stated the conditions (either giving aqueous or ethanol but not both).

QUESTION 08

8.1 Many students knew that they needed to ensure P did not evaporate; the most common error was stating that the tube should be sealed. About a third of students correctly realised the need to cool the test tube.

8.2 Reagents and colour change were well known. Half of all students correctly identified the practical procedure and scored full marks.

8.3 The majority of students knew that there would be effervescence. Just over 50% of the students were able to correctly write the equation; a common error was to show O–Na as covalent bond.

8.4 A third of students correctly identified the product and scored maximum marks on this question. The most common error was incorrect numbering of the carbon chain and showing 3-methylbutanoic acid. Students must state the colour and physical state of the product – red precipitate.

8.5 Just over half of the students answered this question well; several students did not give the structure in the format asked for in the question i.e. displayed.

QUESTION 9

9.1 The majority of students answered the question well. The most common error was not following the ratio in the equation.

9.2 Students know how to do this type of calculation, and two thirds of students answered well. Errors with K_c were either not using [], or missing the 2 on $[X_2]$. A calculation error tended to be forgetting to square [Y].

9.3 The majority of students scored the first mark. Only the more able students were able to give a good explanation as to why the equilibrium changes.

9.4 Just over half of all students were able to score both marks.

9.5 Two thirds of students answered this well. The most common error was incorrect units.